Name: YANG YANG

Tel: (+86) 183 8679 7248 | yangyang@mail.scuec.edu.cn | Homepage

Address: No.182 Minyuan Road, Hongshan District, Wuhan, China

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Education

South-Central Minzu University — Wuhan, China
M.S. in Computer Technology — *Sep 2024 – Jun 2027 (expected)*

South-Central Minzu University — Wuhan, China
B.E. in Network Engineering — *Sep 2020 – Jun 2024* | *GPA: 3.17/4.0*

Research Interests

Medical Image Segmentation, Multimodal Learning, Semi-Supervised Learning

Publications

Journal Papers

Yang, Y., Hong, Y., Yu, W., et al. HSFPN-Det: An Effective Model for Detecting Rice Pests and Diseases. *The Visual Computer (TVC)*, 2025. [Paper]. (**JCR Q2, IF: 3.4**).

Yang, Y., Huang, X., Zhang, X., et al. Learning Symmetric Cross-Modal Representations via Feature Distillation and Contrastive Fusion for Brain Tumor Segmentation. [Submitted to Journal].

Huang, X., **Yang, Y.**, Lv, Y., et al. Ontology-Guided Lesion-Aware Vision-Language Alignment for Retinal Image-Text Retrieval. [Submitted to Journal].

Conference Papers

Yang, Y., Zhu, J., Yang, B., et al. An Improved YOLOv8-based Rice Pest and Disease Detection Method. *Computer Graphics International (CGI)*, 2024. [Paper]. (**CCF C**).

Yang, Y., Huang, X., Zhang, X., et al. Hi-ERMNet: Hierarchical Transformer with Edge Refinement for Retinal Vessel Segmentation. [Under Review].

Papers to be Submitted

Yang, Y., Huang, X., Zhang, X., et al. (2026) Background as Parts: Sub-Background Modeling for Uncertainty-Aware Semi-Supervised Medical Image Segmentation.

Research Experience

Retinal Fine Vessel Segmentation — Apr. 2025 – Sept. 2025
Graduate Researcher, South-Central Minzu University — *Wuhan, China*

- ***Project Title:*** Research on Key Technologies for Multimodal Intelligent Analysis of Retinal Fine Vessels and Disease Prediction
- ***My Responsibilities:*** Used the Sobel operator for edge detection of retinal images and proposed Hi-ERMNet to segment retinal fine vessels. Used Gaussian blur to simulate non-ideal conditions, demonstrating the robustness of Hi-ERMNet.

Wearable Devices and Cardiovascular Disease Analysis — Sept. 2024 – Dec. 2024
Graduate Researcher, South-Central Minzu University — *Wuhan, China*

- ***Project Title:*** Cardiovascular Disease Prediction Based on ECG/PPG Signals from Wearable Devices.
- ***My Responsibilities:*** Completed the preprocessing and analysis of electrocardiogram (ECG) and photoplethysmography (PPG) physiological signals collected by wearable devices. Built a time-series prediction model using Recurrent Neural Networks (RNN) as part of the foundational early-stage process for cardiovascular disease risk prediction.

Evolution of Typical Ship Fire Scenarios Based on Virtual-Real Fusion Jul. 2024 – Nov. 2024
Graduate Researcher, South-Central Minzu University *Wuhan and Shanghai, China*

- ***Project Title:*** Research on Building a Virtual Model of Fire Evolution Based on Data-Mechanism Fusion.
- ***My Responsibilities:*** Used PyroSim to analyze the fire development process and preliminarily proposed a ship fire development model. Applied Physics-Informed Neural Networks (PINNs) to model the ship fire evolution process, supporting early warning and risk assessment of cabin fire scenarios.

Selected Honors & Awards

- First-class Academic Scholarship for Graduate Students Sept. 2025
- Outstanding Graduate Student Sept. 2025
- Second-class Academic Scholarship for Graduate Students Sept. 2024
- National Encouragement Scholarship Sept. 2023

Skills and Software

Programming: Python, C/C++, Java **Tools:** Git, Linux
Scientific Writing: LaTeX **Frameworks:** PyTorch
Data Analysis & Visualization: Numpy, Scipy, Matplotlib

Review Services

- Expert Systems With Applications
- Applied Soft Computing
- Journal of Computer-Aided Design & Computer Graphics
- The Visual Computer
- ACM MM
- Computer Graphics International

Languages

Chinese Mandarin (native), English (CET6 504)